Time Series Ex.2

1. It was determined that a quadratic model was the best fit of the trend for 12 years of data. The data are

22, 18, 21, 19, 17, 16 ,18 ,21, 20, 23, 22, 25

The results from running a multiple regression on these data arc as follows:

corr(Y, t) = .4839 b_0 = 22.95456 b_1 = -1.8317

t-test for b_1 = -3.2301

corr(Y,t^2)= .6307 b_2 =0.1683

t-test for b_2 = 3.9642

R^2 = 0.72 R^2_{adj} = 0.6592 s_e = 1.5513

F-test = 11.6362

Durbin-Watson = 2.3913

Correction factor for the 13th time period = 1.438

Error terms for the model:

t	e_t	t	e_t	t	e_t
1	0.7088	5	- 1.004	9	-0.1044
2	-1.9645	6	- 2.024	10	1.529
3	2.0254	7	-0.381	11	- 1.1743
4	0.6788	8	1.9256	12	-0.9143

(a) What is the equation for trend?

(b) Use the Durbin-Watson statistic to test whether the residuals are correlated or not?

(c) If appropriate, determine the statistical fitness of the model.

(d) Find and interpret Theil's U.

(e) Was the quadratic a good choice to model the trend in the data?

(f) Find the appropriate forecast (point and/or interval) for the 13th time period.

2. For each of the following sets of data, determine whether a model can be constructed either as a linear trend or a curvilinear trend model by using the following steps:

i. Plot the time-series data versus time. Plot first differences of your data. If necessary, plot second differences. Based upon your graphs, what model, either linear or curvilinear, would be appropriate? Explain.

ii. Calculate the correlation coefficient between the time-series variable and time. Test for statistical significance.

iii. If your choice is linear, run a simple bivariate regression against time. Following the sections outlined in this chapter, test the model for the following: Durbin-Watson statistic; statistical significance *of* the slope coefficient; statistical significance *of* the overall regression. Determine *if* the forecast errors are norma11ydistributed. Calculate the forecast summary statistics, MAE, MAPE, R~SE, and Theil's U.

iv. If your choice is nonlinear, justify which of the nonlinear specifications (first differences, exponential, logistic, quadratic, or cubic) is appropriate. Your justification should include the appropriate statistical tests of the model, and calculation of the forecast summary statistics (see EX.1).

v. Using the correct model (either linear or nonlinear), forecast the next three time periods. Construct the 95 percent prediction interval for each forecast. Graph your results: actual values, forecast values, and the 95 percent prediction interval.

(a). The following is revenue data, in millions of dollars, for 1981-1990 from Springs Inc., a textile manufacturer in Fort Mill, South Carolina:

Year	Sales
81	917.0
82	874.5
83	894.4
84	945.0
85	1,013.5
86	1,505.0
87	1,661.1
88	1,824.8
89	1,909.3
90	1,878.0

(b). The following table presents the total fish catch (in millions of pounds) in the United States:

Year	Fish
1972	4,806
1973	4,858
1974	4,967
1975	4,877
1976	5,388
1977	5,271
1978	6,028
1979	6,267
1980	6,482
1981	5,977
1982	6,367
1983	6,439
1984	6,438
1985	6,258
1986	6,031
1987	6,896
1988	7,192

3. Column 2 of the attached table is the yearly tobacco production of a certain country from 1980- 1991.

Year	Tab. Prodn	SES forecast	MA3 forecast
80	2315		
81	2344		
82	2228		
83	1855		
84	1887		
85	1968		
86	1710	1787	1903.33
87	1804		
88	1906		
89	1705		
90	1749		
91	1742		

We want to compare the forecasting performance of two methods, simple exponential smoothing (SES) and 3-points moving average(MA3). Forecast values for the year 1986 were given.

(a) Using the smoothing constant 0.8, calculate SES forecasts for 1987 to 1991.

(b) Calculate MA3 forecasts for 1987 to 1991 .

(c) Calculate the SSE (total sum of squares for forecast errors) for both methods during 1986- 1991 and hence conclude which method is better.

4. Consider a linear trend process,

$$y_t = b_1 + b_2 t + \varepsilon_t, \text{where } \varepsilon_t \sim N(0, \sigma^2).$$

Show that the smoothed statistics of double exponential smoothing, S_T & $S_T^{[2]}$, satisfy the following relationship:

$$E(S_T^{[2]}) = E(S_T) - \frac{1-\alpha}{\alpha} b_2 .$$